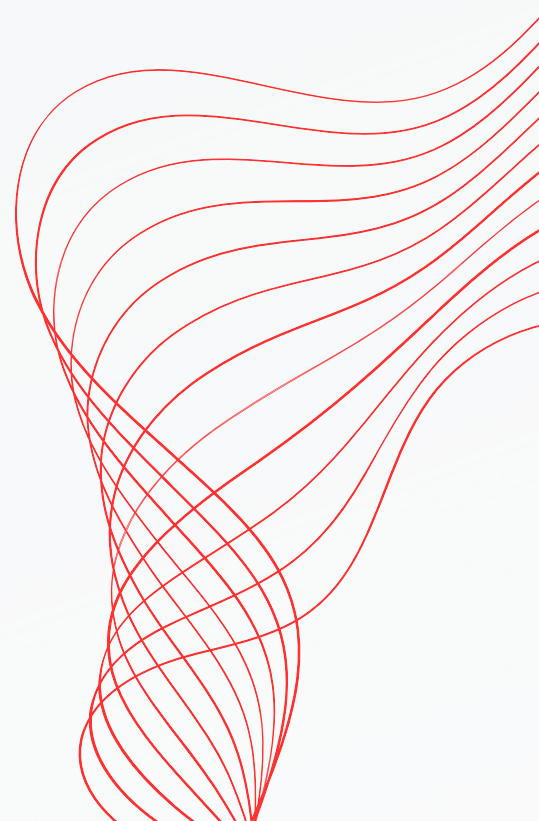
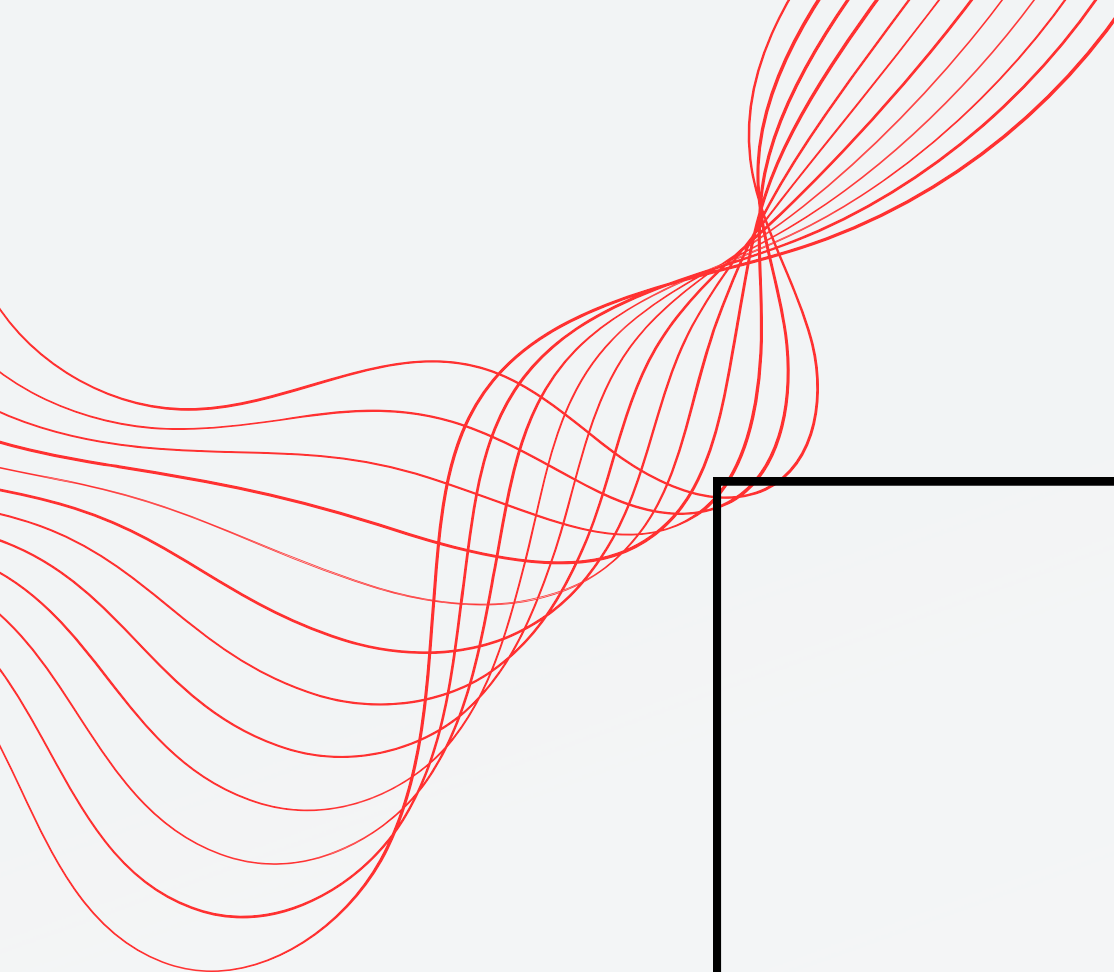




# **ENGG2112 PROJECT**

# **BUSHFIRE**

# **DETECTION**

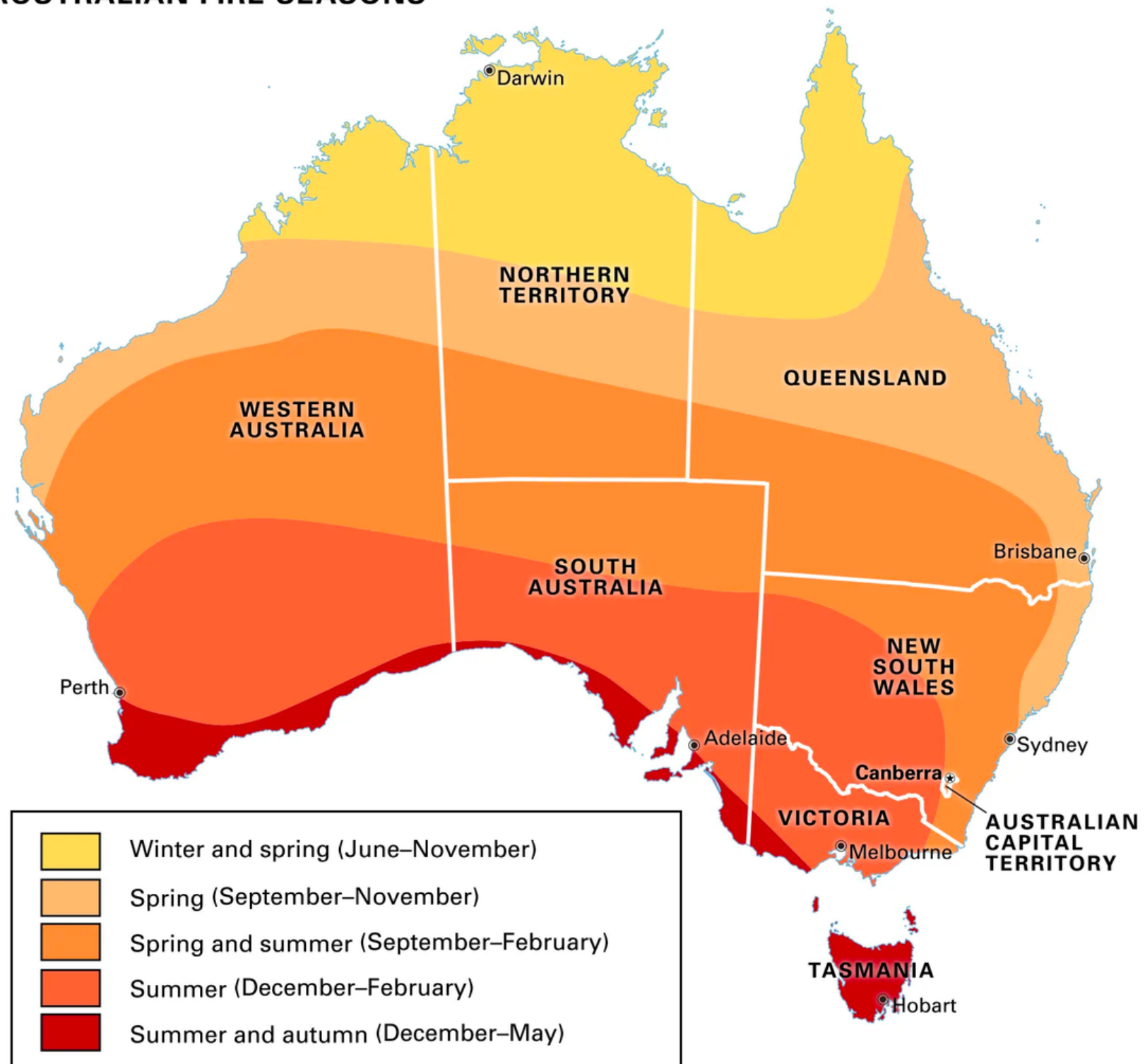
.

# CONTENT

- 01** STATISTICS
- 02** GOALS AND OBJECTIVES
- 03** DATA COLLECTION
- 04** DATA CLEANING
- 05** MACHINE LEARNING METHODS
- 06** BEST METHODS
- 07** TARGET AUDIENCE & RECOMMENDATIONS
- 08** FUTURE DEVELOPMENT

# STATISTICALLY

AUSTRALIAN FIRE SEASONS



**45,000 - 60,000  
BUSHFIRES IN  
AUSTRALIA EACH YEAR.**





## DEATHS

Between 1967 to 2013 bushfires have resulted in **8000** injuries and **433** deaths



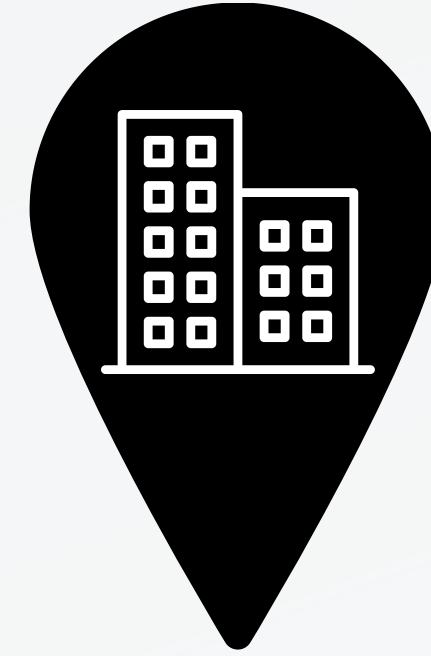
## DESTRUCTION

Over **19 million hectares** have been burnt during one of Australia's most catastrophic "Black Summer" bushfire peak seasons from 2019 to 2020



## WILDLIFE

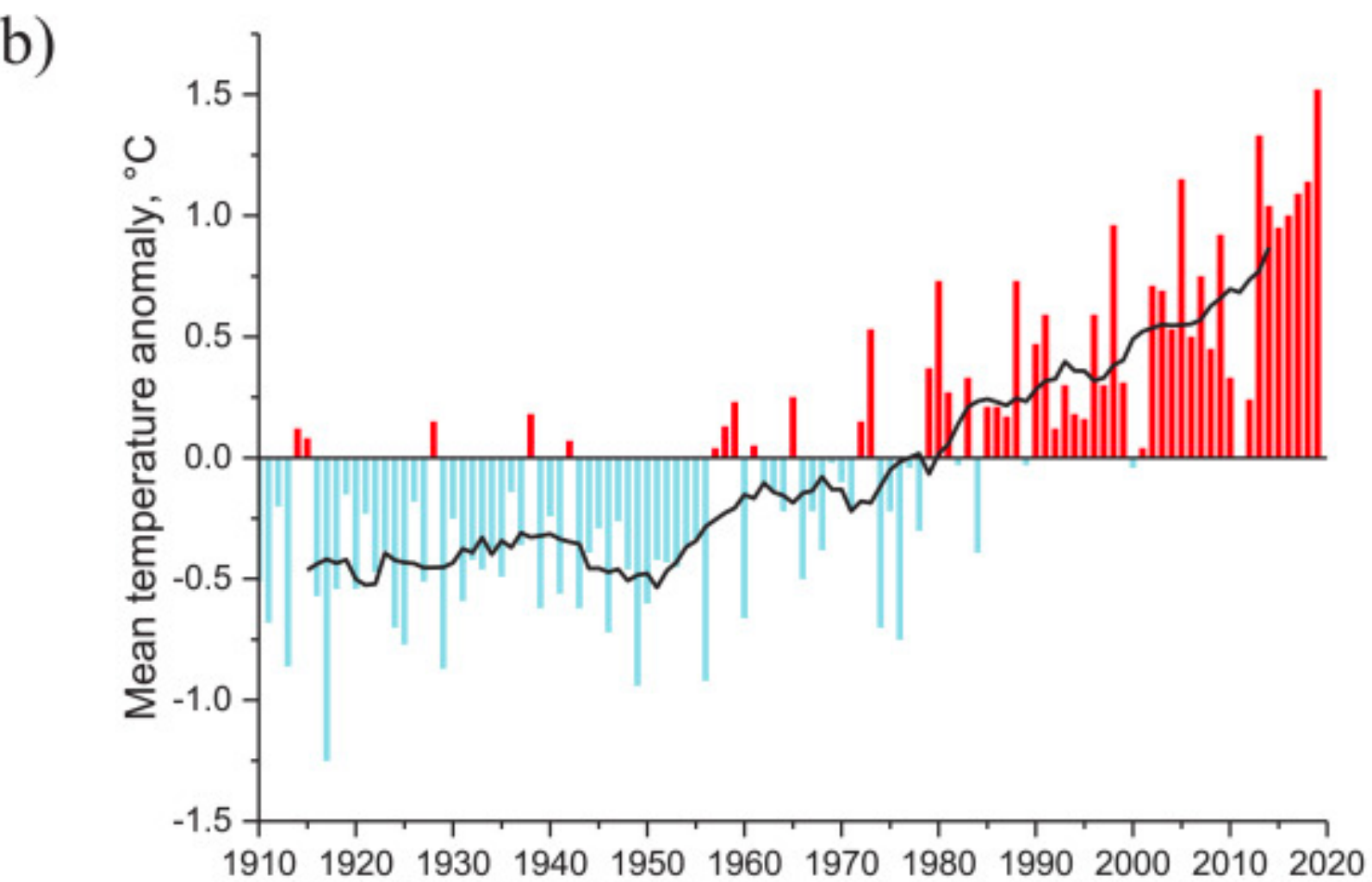
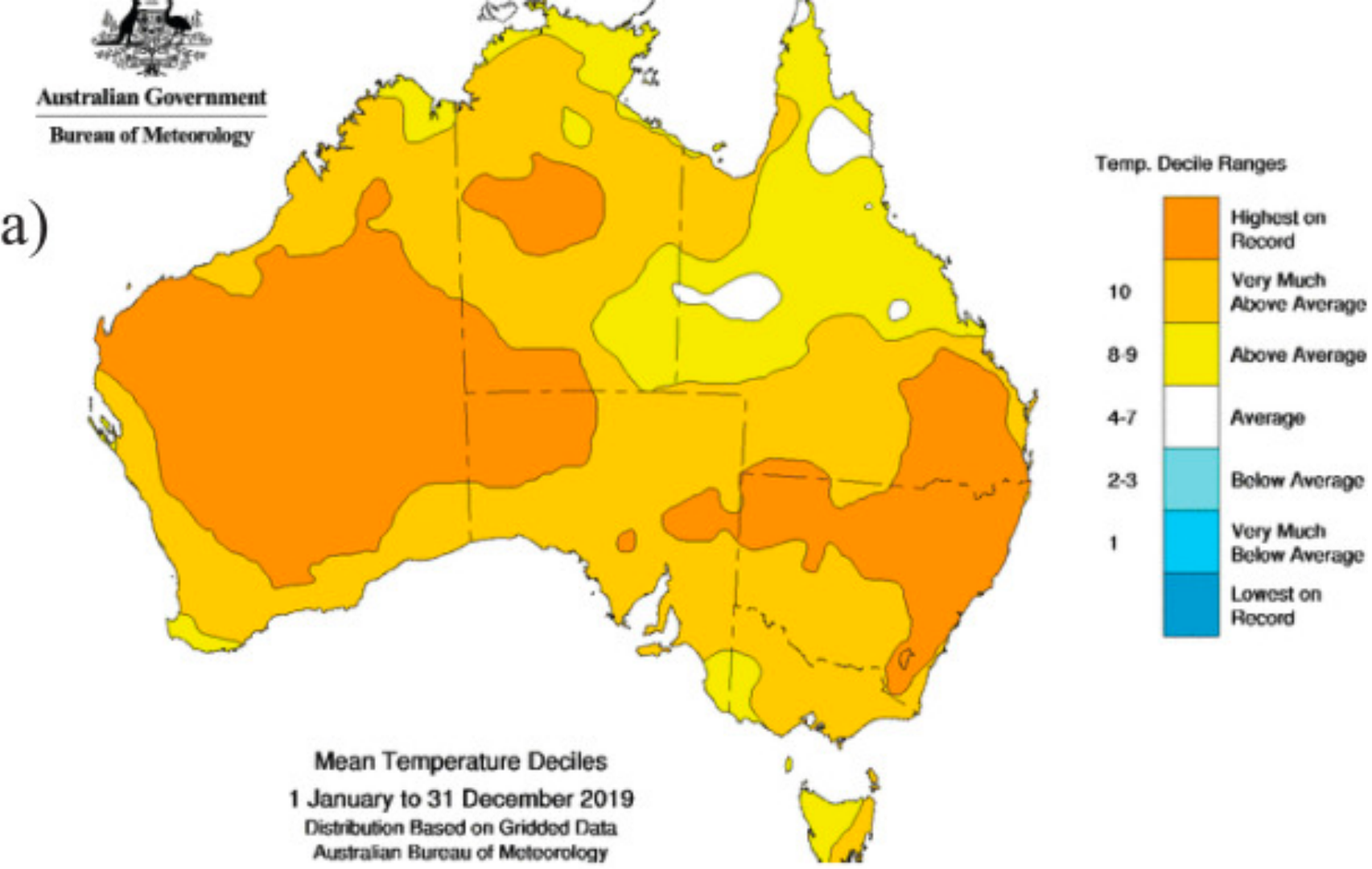
Killing of more than **1.25 billion** animals.



## DEMOLITION

Households and commercial buildings were partially damaged or destroyed, pushing early insurance claims to a whopping value of **1.9 billion** Australian dollars in February 2020.





# Increasing Prevelance

# GOALS AND OBJECTIVES

## Prediction

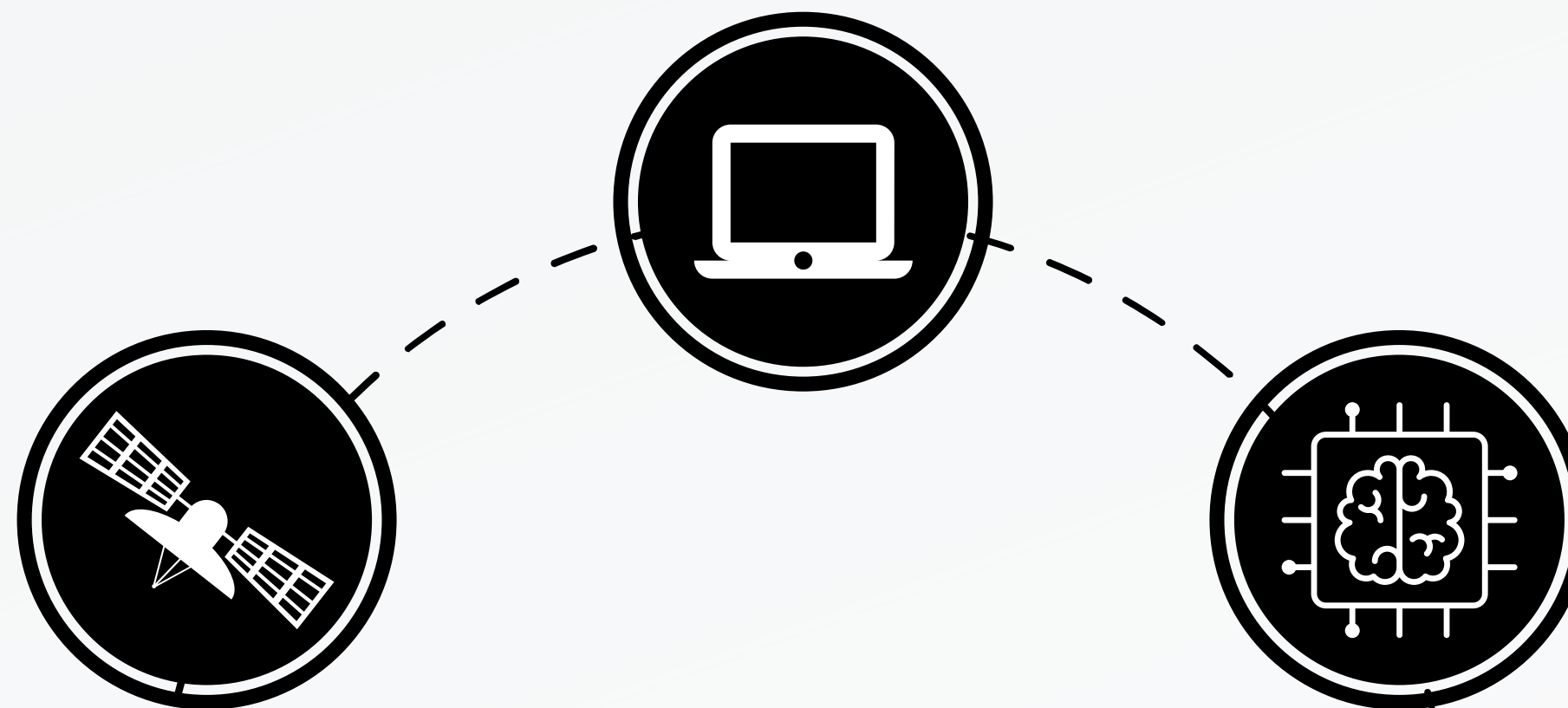
By using machine learning we aim to detect bushfires with a high confidence level.

## Detection and Reporting

If confidence is greater than 90% trigger and alert and report bushfire to local government and emergency services

## Monitor

Continue to monitor bushfires and if greater risk is generated to continue to report.



# DATA COLLECTION

How did we get our data?

Our data was collected by 2 NASA satellites, MODIS and VIIRS which launched in 1999 and 2002 respectively

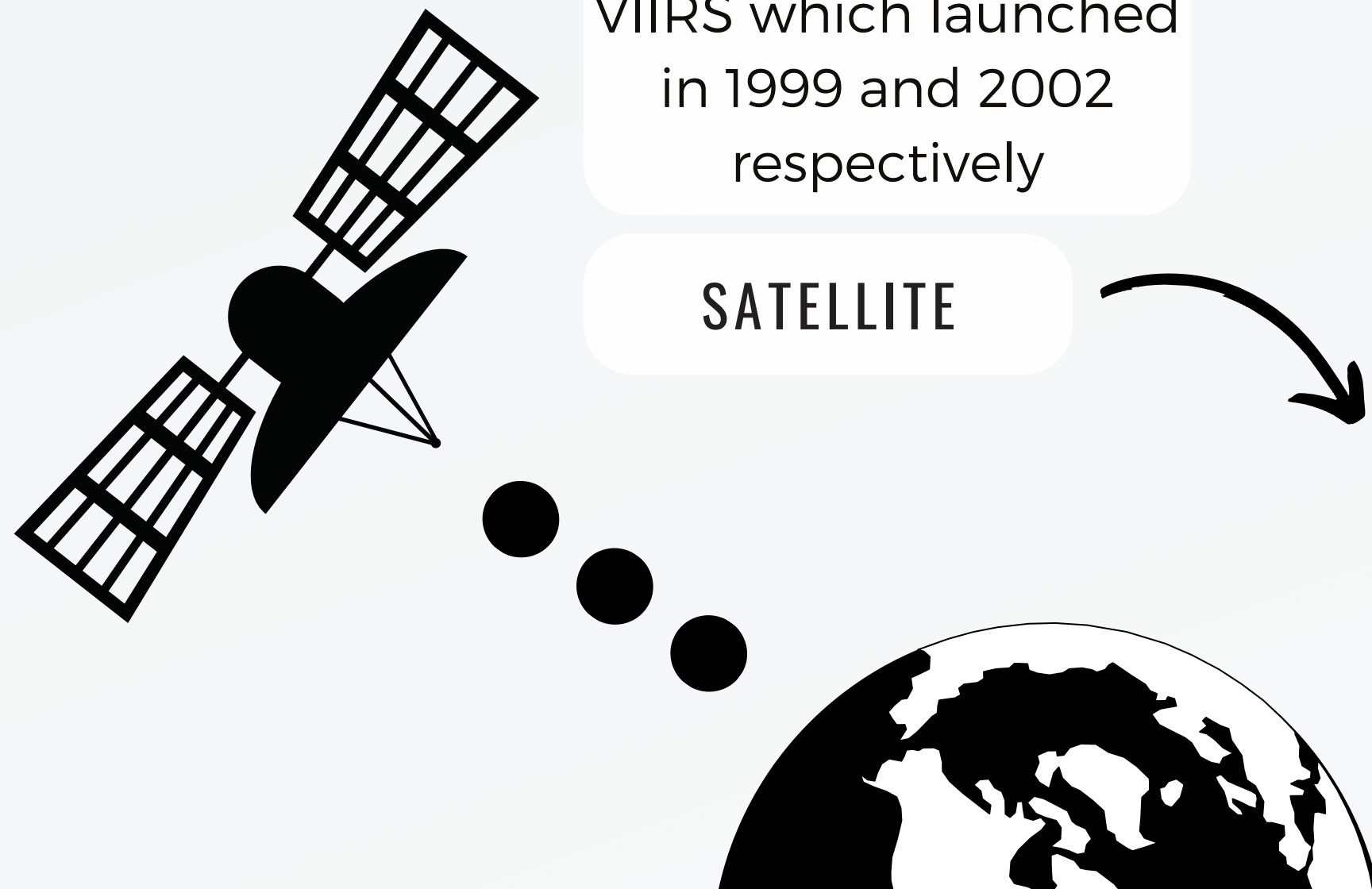
SATELLITE

Using Advanced Spaceborne Thermal Emission and Reflection Radiometer (ASTER), the Clouds and Earth's Radiant Energy System (CERES), the Multi-angle Imaging SpectroRadiometer (MISR), the Measurement of Pollution in the Troposphere (MOPITT), and the Moderate Resolution Imaging Spectroradiometer

DATA COLLECTION

The data from 1 Sept, 2019 to 31 Jan, 2020. was the uploaded for public access on KAGGLE

KAGGLE



# DATASETS

(CLASSIFIED BASED ON THE SATELLITE)

## V1 DATASET (VIIRS)

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 1119683 entries, 0 to 1013969
Data columns (total 15 columns):
#   Column          Non-Null Count  Dtype
---  -
0   latitude        1119683 non-null float64
1   longitude       1119683 non-null float64
2   bright_ti4      1119683 non-null float64
3   scan            1119683 non-null float64
4   track           1119683 non-null float64
5   acq_date        1119683 non-null object
6   acq_time        1119683 non-null int64
7   satellite       1119683 non-null object
8   instrument      1119683 non-null object
9   confidence      1119683 non-null object
10  version         1119683 non-null object
11  bright_ti5      1119683 non-null float64
12  frp             1119683 non-null float64
13  type            105713 non-null float64
14  daynight        1013970 non-null object
dtypes: float64(8), int64(1), object(6)
memory usage: 136.7+ MB
```

## M6 DATASET (MODIS)

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 183163 entries, 0 to 34855
Data columns (total 15 columns):
#   Column          Non-Null Count  Dtype
---  -
0   latitude        183163 non-null float64
1   longitude       183163 non-null float64
2   brightness      183163 non-null float64
3   scan            183163 non-null float64
4   track           183163 non-null float64
5   acq_date        183163 non-null object
6   acq_time        183163 non-null int64
7   satellite       183163 non-null object
8   instrument      183163 non-null object
9   confidence      183163 non-null int64
10  version         183163 non-null object
11  bright_t31      183163 non-null float64
12  frp             183163 non-null float64
13  daynight        183163 non-null object
14  type            148307 non-null float64
dtypes: float64(8), int64(2), object(5)
memory usage: 22.4+ MB
```



# DATA CLEANING

Dropping null values  
from both the tables

Mapping the  
'confidence' values of V1  
dataset - 'l', 'n', 'h', to 0, 1,  
2  
and finding the  
quartiles for  
'confidence' values of  
M6 datasets and  
mapping it to 0, 1, 2 for  
uniformity

Made correlation table  
to see how an attribute  
affects other features

Used geopandas to  
visualise the data points  
using latitudes and  
longitudes

**Two different  
datasets have  
been maintained  
since the pixel  
size is different  
for both the  
datasets - 1km  
radius for M6 and  
375m radius for V1**

# V1 DATASET (VIIRS)

```
<ipython-input-40-7d12d379e637>:1: FutureWarning: The default value of numeric_only in DataFrame.corr is deprecated. In a future version,
corr_V1 = df_V1.corr()
```

|                    | latitude  | longitude | bright_ti4 | scan      | track     | acq_time  | bright_ti5 | frp       | type      | confidence_encoded |
|--------------------|-----------|-----------|------------|-----------|-----------|-----------|------------|-----------|-----------|--------------------|
| latitude           | 1.000000  | -0.404740 | 0.142739   | 0.015063  | 0.058664  | -0.227170 | 0.243828   | -0.064744 | -0.036457 | -0.073179          |
| longitude          | -0.404740 | 1.000000  | -0.199063  | -0.040432 | -0.126458 | 0.104951  | -0.299144  | -0.054198 | -0.026737 | 0.008510           |
| bright_ti4         | 0.142739  | -0.199063 | 1.000000   | -0.040779 | -0.021412 | -0.725574 | 0.677341   | 0.290829  | -0.037586 | 0.255378           |
| scan               | 0.015063  | -0.040432 | -0.040779  | 1.000000  | 0.530126  | -0.017889 | -0.074358  | 0.023675  | -0.002685 | 0.007026           |
| track              | 0.058664  | -0.126458 | -0.021412  | 0.530126  | 1.000000  | -0.074794 | -0.123391  | -0.013216 | -0.009307 | 0.141257           |
| acq_time           | -0.227170 | 0.104951  | -0.725574  | -0.017889 | -0.074794 | 1.000000  | -0.632984  | -0.222908 | 0.032020  | -0.020269          |
| bright_ti5         | 0.243828  | -0.299144 | 0.677341   | -0.074358 | -0.123391 | -0.632984 | 1.000000   | 0.435116  | -0.038739 | 0.042396           |
| frp                | -0.064744 | -0.054198 | 0.290829   | 0.023675  | -0.013216 | -0.222908 | 0.435116   | 1.000000  | -0.012472 | 0.122775           |
| type               | -0.036457 | -0.026737 | -0.037586  | -0.002685 | -0.009307 | 0.032020  | -0.038739  | -0.012472 | 1.000000  | -0.002057          |
| confidence_encoded | -0.073179 | 0.008510  | 0.255378   | 0.007026  | 0.141257  | -0.020269 | 0.042396   | 0.122775  | -0.002057 | 1.000000           |

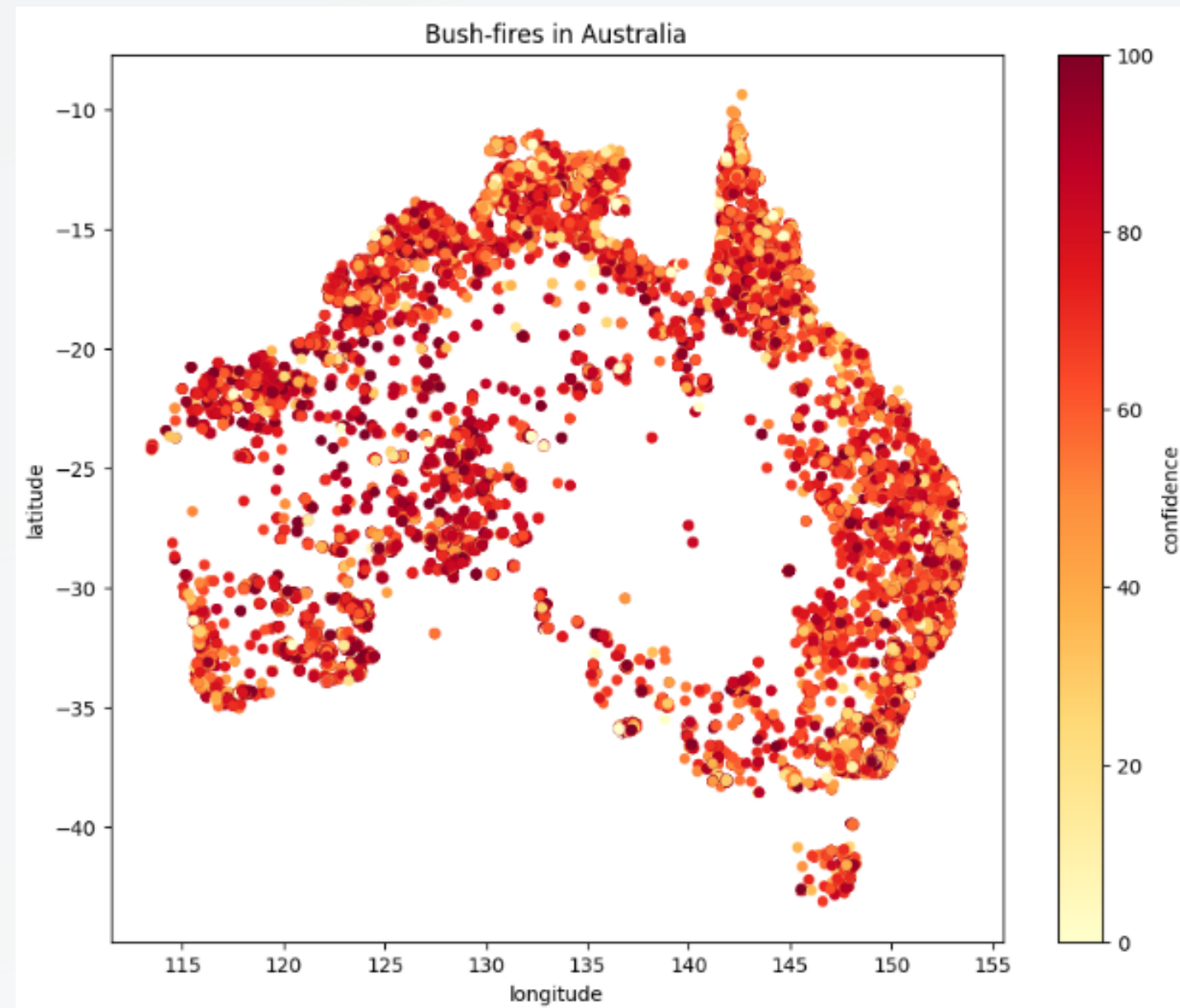
# M6 DATASET (MODIS)

```
<ipython-input-41-c8d92452d8b8>:1: FutureWarning: The default value of numeric_only in DataFrame.corr is deprecated. In a future version, it will de
corr_M6 = df_M6.corr()
```

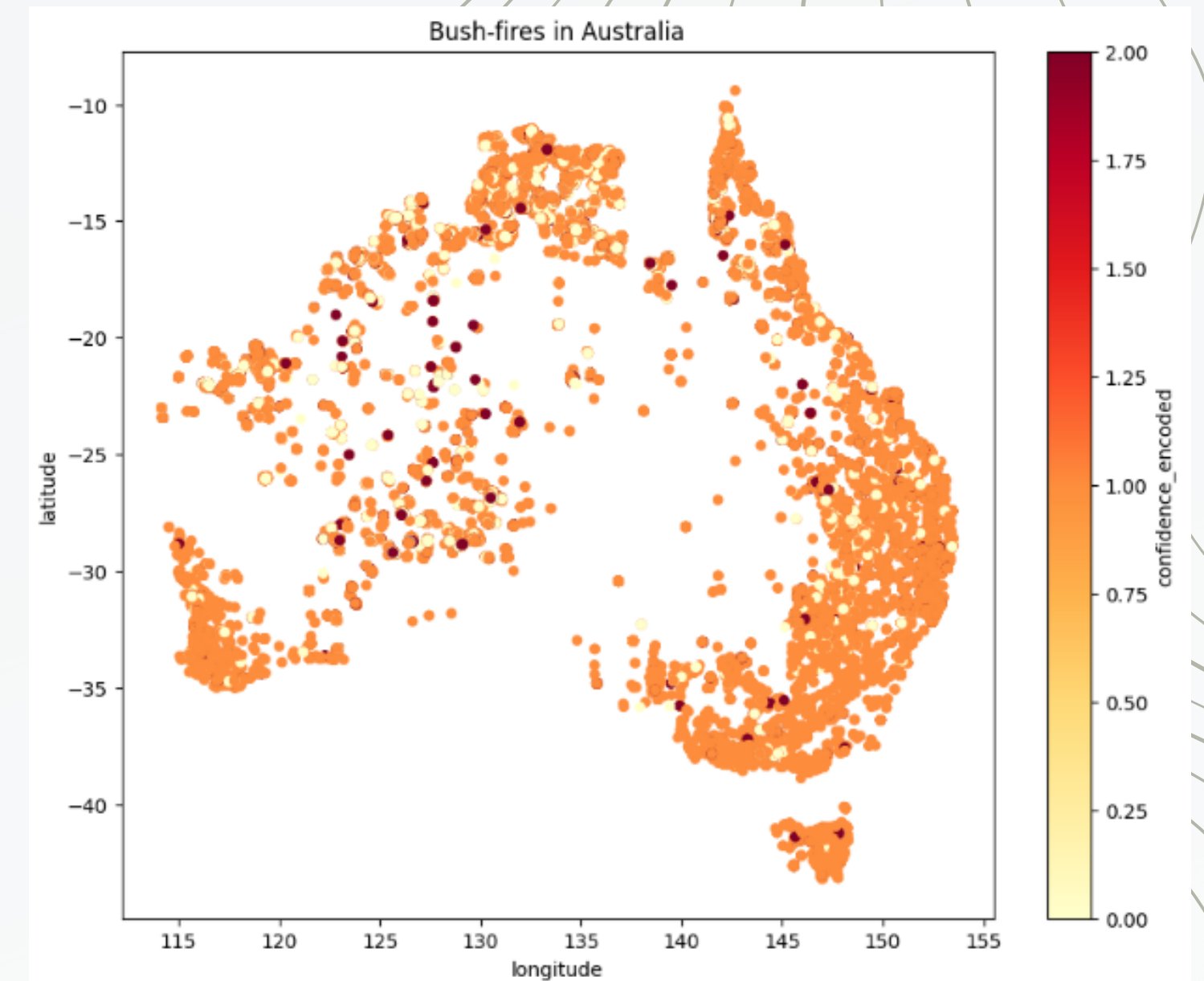
|                    | latitude  | longitude | brightness | scan      | track     | acq_time  | confidence | bright_t31 | frp       | type      | confidence_encoded |
|--------------------|-----------|-----------|------------|-----------|-----------|-----------|------------|------------|-----------|-----------|--------------------|
| latitude           | 1.000000  | -0.410281 | -0.096068  | 0.076601  | 0.081531  | -0.235878 | -0.066562  | 0.168618   | -0.097161 | -0.017629 | -0.168513          |
| longitude          | -0.410281 | 1.000000  | -0.107625  | -0.121951 | -0.127085 | 0.181447  | -0.052374  | -0.305903  | -0.064572 | -0.033967 | 0.000972           |
| brightness         | -0.096068 | -0.107625 | 1.000000   | -0.106967 | -0.114748 | -0.302752 | 0.531152   | 0.684595   | 0.631259  | -0.029414 | 0.604006           |
| scan               | 0.076601  | -0.121951 | -0.106967  | 1.000000  | 0.982930  | -0.018234 | -0.102375  | -0.099329  | 0.183865  | -0.012571 | -0.121336          |
| track              | 0.081531  | -0.127085 | -0.114748  | 0.982930  | 1.000000  | -0.010770 | -0.107795  | -0.102119  | 0.178375  | -0.014591 | -0.127590          |
| acq_time           | -0.235878 | 0.181447  | -0.302752  | -0.018234 | -0.010770 | 1.000000  | -0.058251  | -0.448394  | -0.082515 | -0.013056 | 0.052968           |
| confidence         | -0.066562 | -0.052374 | 0.531152   | -0.102375 | -0.107795 | -0.058251 | 1.000000   | 0.270612   | 0.246908  | -0.035219 | 0.788719           |
| bright_t31         | 0.168618  | -0.305903 | 0.684595   | -0.099329 | -0.102119 | -0.448394 | 0.270612   | 1.000000   | 0.375173  | -0.009806 | 0.221567           |
| frp                | -0.097161 | -0.064572 | 0.631259   | 0.183865  | 0.178375  | -0.082515 | 0.246908   | 0.375173   | 1.000000  | -0.014088 | 0.322738           |
| type               | -0.017629 | -0.033967 | -0.029414  | -0.012571 | -0.014591 | -0.013056 | -0.035219  | -0.009806  | -0.014088 | 1.000000  | -0.028311          |
| confidence_encoded | -0.168513 | 0.000972  | 0.604006   | -0.121336 | -0.127590 | 0.052968  | 0.788719   | 0.221567   | 0.322738  | -0.028311 | 1.000000           |

# VISUALISATION OF DATA ON MAP

**LOCATIONS OF BRIGHT PIXEL SPOTS  
DETECTED BY THE SATELLITE - MODIS  
(ACCORDING TO CONFIDENCE LEVELS)**



**LOCATIONS OF BRIGHT PIXEL SPOTS  
DETECTED BY THE SATELLITE - VIIRS  
(ACCORDING TO CONFIDENCE LEVELS)**





# MACHINE LEARNING METHODS

- LINEAR REGRESSION
- KNN CLASSIFICATION
- KNN REGRESSION
- MULTI-LAYER PERCEPTRON (MLP)
- NEURAL NETWORKS



# LINEAR REGRESSION

- Appropriate for continuous target variable – 'confidence'
- Its simplicity and interpretability
- Poor performance shows that the model isn't a good fit and data exhibits complex non-linear patterns

|                    | V1   | M6   |
|--------------------|------|------|
| R2_SCORE           | 0.14 | 0.46 |
| MEAN_SQUARED_ERROR | 0.16 | 0.38 |

# KNN CLASSIFICATION

- Ability to handle non-linear relationships of data.
- Makes no assumption about underlying data distribution.
- It doesn't require extensive parameter tuning or complex model training, making it a good choice.
- Extremely good performance shows that the model was able to predict the class labels for the majority portion of the instances correctly.

|                | M6              | V1              |
|----------------|-----------------|-----------------|
| ACCURACY_SCORE | 0.85/<br>85.26% | 0.95/<br>95.78% |
| F1_SCORE       | 0.85/<br>85.09% | 0.95/<br>95.63% |

# KNN REGRESSION

- Can capture non-linear relationships between the input features and the target variable
- Adapts well to different datasets and problem domains
- Performance metrics shows the model has made accurate predictions and is a reasonably good fit as it explains at least 70% of the data correctly with room for improvement.

|          | V1    | M6    |
|----------|-------|-------|
| R2_SCORE | 0.714 | 0.796 |
| MSE      | 0.0   | 0.0   |

# FEEDFORWARD NEURAL NETWORK

- Ability to capture complex and intricate non-linear relationships.
- It can automatically learn hierarchical representation of data with multiple layers.
- Performance table shows it's a good fit for V1 dataset but not for M6 dataset due to low accuracy and higher test loss in comparison.

|               | V1            | M6         |
|---------------|---------------|------------|
| Test accuracy | 0.789         | 0.568      |
| Test loss     | -1535646464.0 | -9235617.0 |



# MULTI-LAYER PERCEPTRON (MLP)

- Effective in learning and representing patterns within the satellite's imagery data.
- Performance table exhibits it as a really good model able to make accurate predictions for majority of the datasets.

|                    | V1              | M6              |
|--------------------|-----------------|-----------------|
| ACCURACY_<br>SCORE | 0.96/<br>96.32% | 0.87/<br>87.72% |

# BEST MODELS



## **KNN Classification**

Accuracy for M6: 85.26%

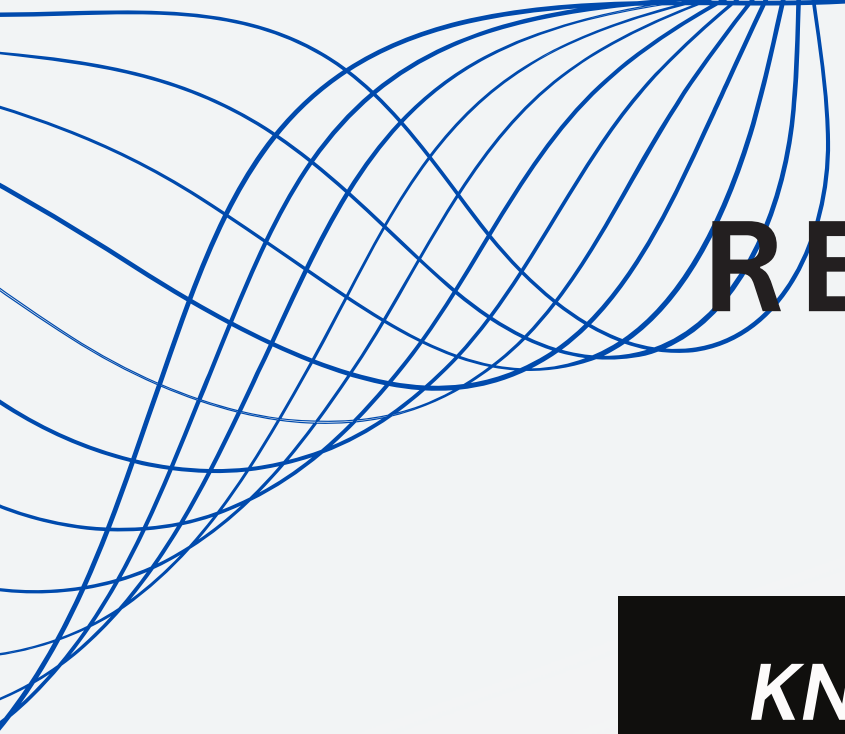
Accuracy for V1: 95.63%



## **Multi-layer Perceptron**

Accuracy for M6: 87.72%

Accuracy for V1: 96.32%



# REASONS FOR PICKING THE BEST MODELS

## ***KNN CLASSIFICATION***

**Effective Handling of Local Variations**

**Proximity-based Classification**

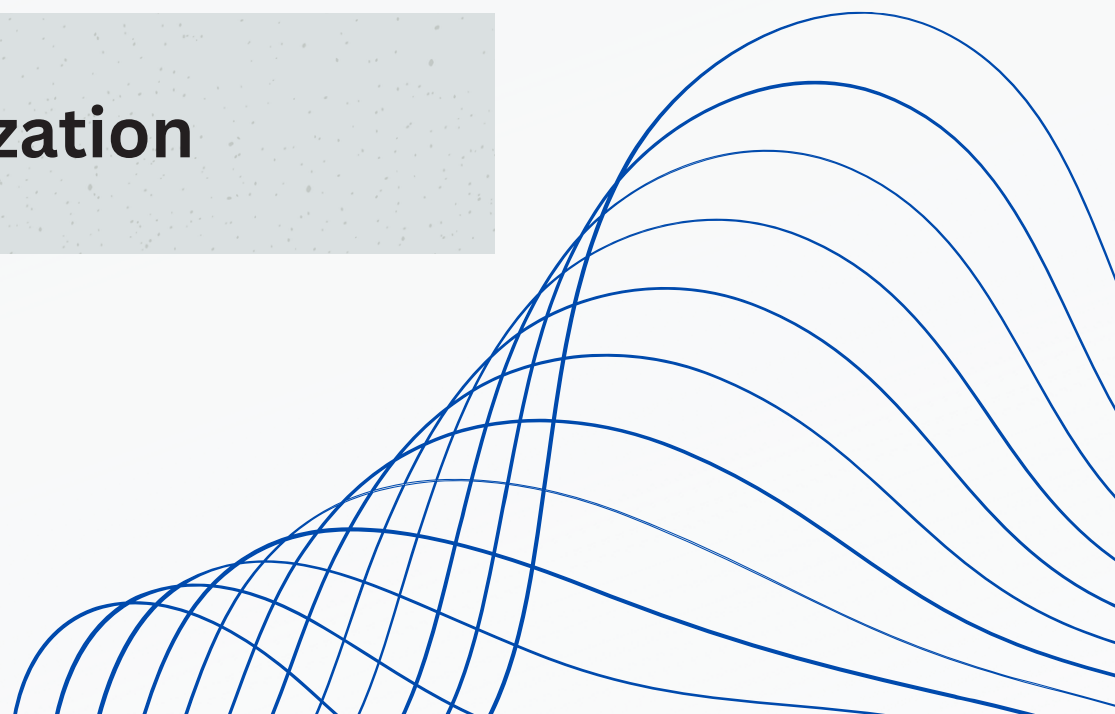
**Flexibility and Scalability**

## ***MULTI-LAYER PERCEPTRON***

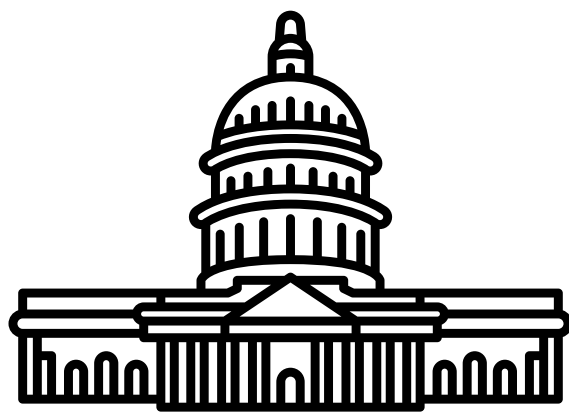
**Nonlinear Mapping**

**Feature Extraction**

**Generalization**

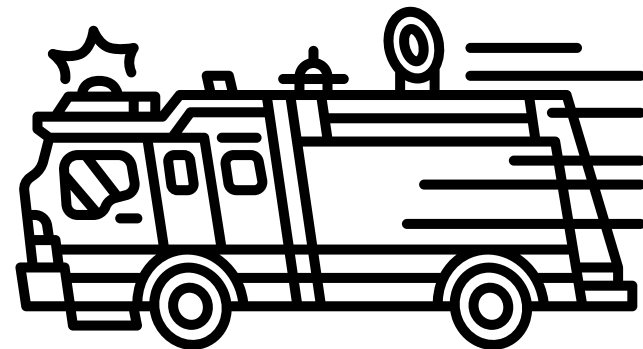


# TARGET AUDIENCE



## **Government Agencies**

Responsible for policy-making, emergency services, environmental protection, and public safety



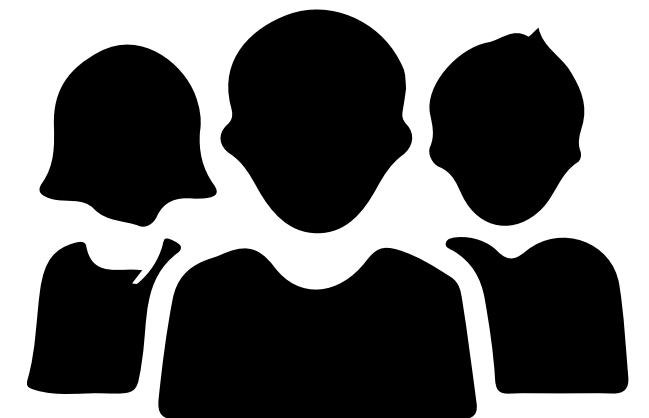
## **Emergency Service Agencies**

Fire departments and organizations involved in managing and responding to bushfire incidents.



## **Technology Companies**

Companies specializing in data analytics, and satellite imagery, offering solutions for bushfire detection.

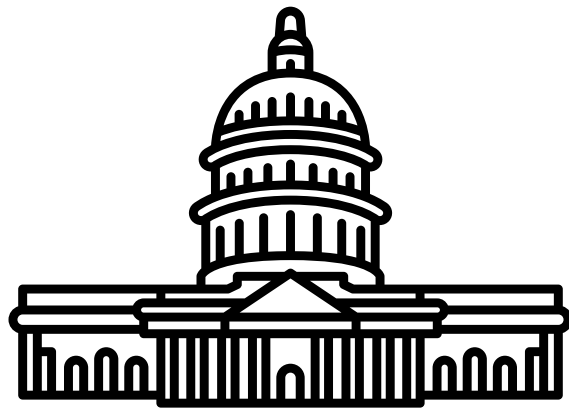


## **General Public**

Individuals who are potentially affected by bush fires



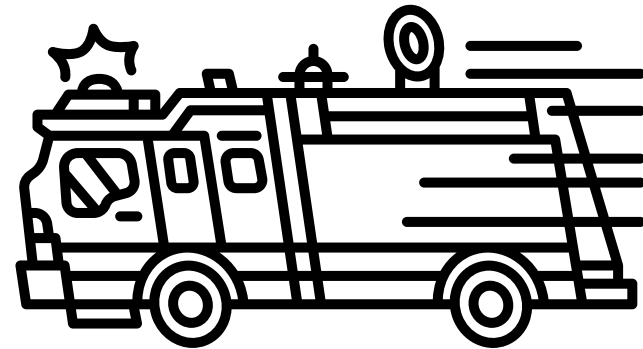
# TARGET AUDIENCE & RECOMMENDATIONS



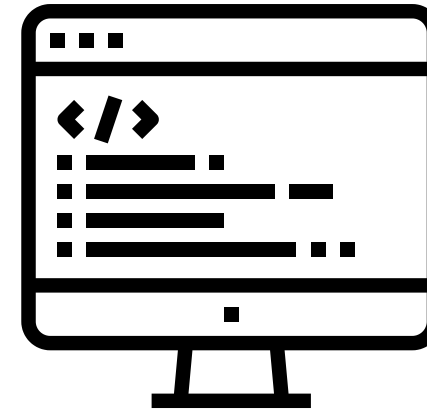
## **Government Agencies**

Can incorporate the ML-based system into decision-making and emergency protocols.

This will lead to faster response, early alerts, and improved the resource mobilization to minimizing impact

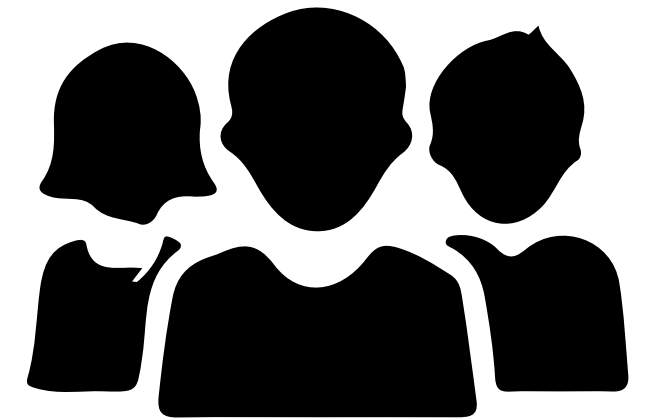


## **Emergency Service Agencies**



## **Technology Companies**

Collaborating with tech companies can help us develop commercial solutions, leading to the improvement of usability and scalability.



## **General Public**

Can help the public stay alert and reducing the risk of loss of life

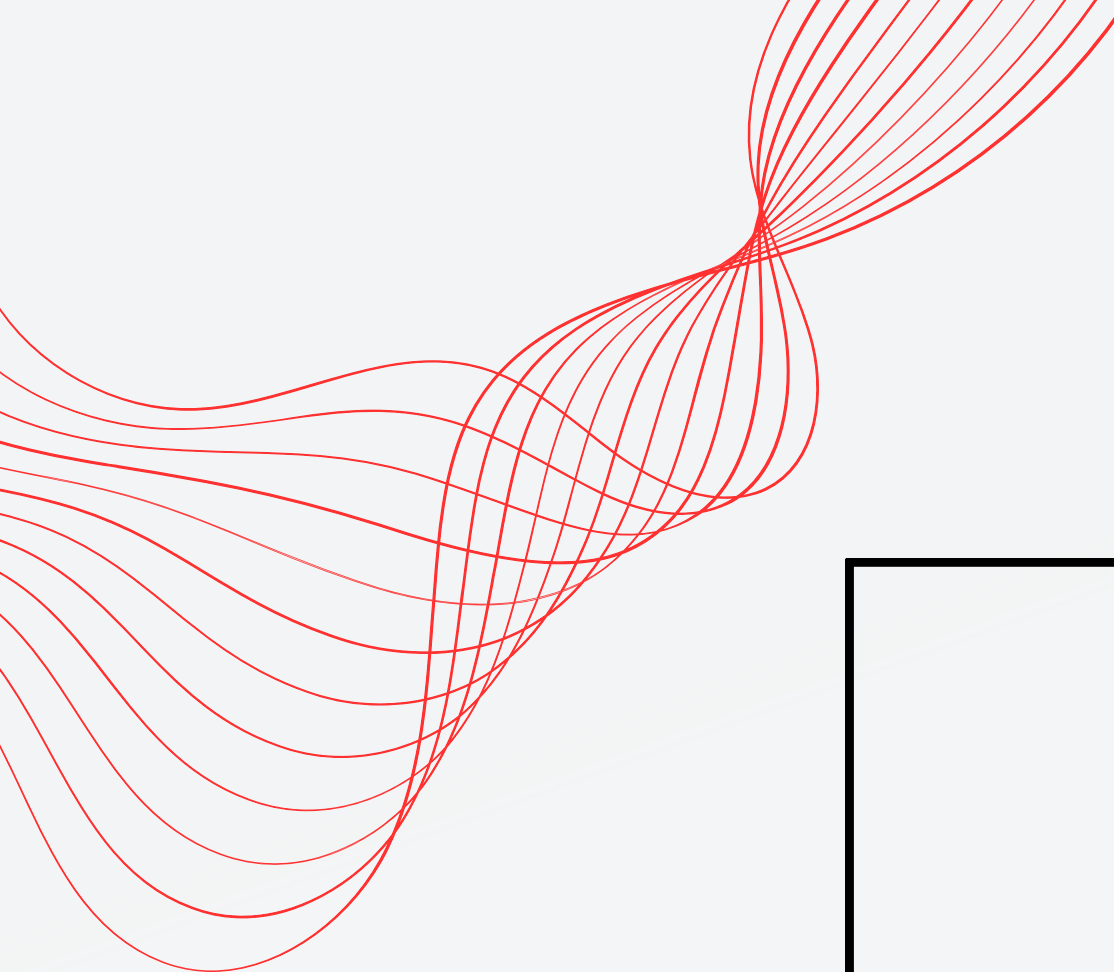
# FUTURE DEVELOPMENTS



Incorporating data such as optical, infrared, and radar imagery, from multiple satellite sources to leverage the strengths of each data type.



Integrate additional relevant data, such as weather conditions, vegetation indices, land cover maps, and historical fire records, into the machine learning models. This can enhance the detection system's understanding and enable more accurate predictions by considering environmental factors.



**THANK  
YOU**

TUT-02 GROUP 10  
SAMIKSHA AVEG (510574673)  
GAYATRI MOTAPARTHI (510671237)  
AIDAN THOMAS (521511398)

